

Retail POS Sales Analysis

Excel Capstone Project — FY 2024-25

Project Report

Author	Dharshini Gopi
Project type	Excel Data Analytics Capstone
Dataset	Sales_Transaction.xlsx (Tally POS export)
Period covered	April 2024 – March 2025 (FY 2024-25)
Tools used	Microsoft Excel, Power Query, PivotTables

This report documents the cleaning, validation, and analysis of one fiscal year of retail point-of-sale transactions, and states plainly what the dataset can and cannot support.

1. Objective

The purpose of this project is to take a raw, unedited Tally point-of-sale export and turn it into a validated dataset that can be trusted for reporting. The project is deliberately framed around the data quality problem rather than around a pre-built dashboard template: the source file contains structural errors serious enough that most of its derived columns cannot be used as they are, and identifying that is the core analytical task.

A secondary objective is to extract every finding the cleaned data can honestly support — transaction value distribution, verified time patterns, and a proof that no revenue was lost or duplicated during cleaning — while explicitly scoping out analyses the data cannot support, such as product mix, margin, or customer segmentation.

2. Dataset Overview

The source file, **Sales_Transaction.xlsx**, is a single-sheet export from Tally covering one voucher type (TSP POS) across the 2024-25 fiscal year.

Property	Value
Raw rows	3,413
Valid transactions after cleaning	3,387
Voucher series	TSP/001\24-25 to TSP/3388\24-25
Voucher type	TSP POS (single type throughout)
Fiscal period	April 2024 – March 2025
Total revenue (cleaned)	Rs. 2,92,30,332.71
Columns in source	Date, TxnDate, Vch Type, Vch No., Amount, Month, Year, Quarter

3. Data Quality Findings

Six distinct issues were identified in the raw export. Each was diagnosed before any cleaning step was applied, so that the fix addressed the actual cause rather than a symptom.

3.1 Stacked exports, not a single table

The file is a set of monthly exports pasted end to end. Header rows (“Vch Type”, “Amount”, “Debit”) appear inside the data body, and 10 block subtotal rows sit in the Amount column alongside genuine transactions — values ranging from Rs. 17.4 lakh to Rs. 40.3 lakh that would otherwise be misread as individual sales.

3.2 The Date column is unreliable after row 266

Vouchers TSP/001 through TSP/266 carry genuine transaction dates spread evenly across the fiscal year. Every row from TSP/267 onward is stamped with the single date 31 March 2025, a fill-down artifact rather than a real transaction date. Only 7.9% of transactions carry a trustworthy date.

3.3 Month / Year / Quarter columns disagree with Date

These derived columns conflict with the Date column in 3,075 of 3,389 rows (91%). Transactions dated April 2024 are labelled “January 2024”, and the mismatch is not a fixed offset — it drifts across the file. The Year column also contains the value 2026, outside the fiscal year the file claims to cover. The Quarter column applies calendar-quarter logic to Indian fiscal-year data, making it incorrect independently of the date mismatch.

3.4 Amount stored as text

The Amount column mixes numeric values with the literal strings “Debit” and “Amount”, plus 14 blank cells, requiring type coercion before any aggregation could be performed.

3.5 TxnDate is fully redundant

Date and TxnDate are identical in all 3,413 rows with no exceptions. TxnDate was removed from the cleaned dataset.

3.6 Duplicate and blank voucher numbers

25 duplicated values were found in Vch No., concentrated in the 24 structural blank rows that separate one month's export block from the next.

4. Methodology

Cleaning and analysis followed a fixed sequence so that every downstream figure can be traced back to a specific transformation step.

- **Import** — loaded the raw export into Power Query without modification.
- **Isolate structural rows** — flagged and removed embedded header rows and block subtotal rows before they could contaminate the transaction-level data.
- **Type correction** — coerced Amount to a numeric type, resolving the text values and blanks.
- **Column rationalisation** — dropped TxnDate (fully redundant) and the unreliable Month, Year and Quarter columns, retaining only fields that could be independently verified.
- **Deduplication** — resolved duplicate and blank voucher numbers tied to structural blank rows.
- **Validation** — reconciled the sum of the 10 extracted subtotal rows against the sum of the cleaned transaction rows.
- **Analysis** — computed distribution statistics on the full cleaned dataset and time-based patterns on the subset of transactions with verified dates only.

5. Validation

The cleaned dataset is provably complete. The 10 subtotal rows extracted from the raw file sum to **Rs. 2,92,30,332.71**, which matches the sum of the 3,387 cleaned transaction rows to the paisa. No revenue was lost or double-counted during the cleaning process.

6. Analysis and Findings

6.1 Transaction value distribution (all 3,387 transactions)

Metric	Value
Mean	Rs. 8,630
Median	Rs. 7,380
Minimum	Rs. 400
90th percentile	Rs. 15,172
99th percentile	Rs. 32,814
Maximum	Rs. 47,320

Mean sits above median, indicating a right-skewed distribution: a small number of higher-value sales pull the average upward while typical transactions cluster closer to Rs. 7,000–8,000. The median is therefore the more representative measure of a “typical” sale.

6.2 Time patterns (verified-date subset only, n = 266)

Monthly transaction counts within the date-verified subset are relatively stable, ranging from 16 transactions in July 2024 to 26 in April, September and February. Because this subset represents only

7.9% of all transactions and was not a random sample — it is simply the portion of the file where the date field happened to remain intact — these patterns are reported as **indicative, not conclusive**.

6.3 Scope limitations

The export contains no product, category, customer, quantity, cost, salesperson, or location fields. As a result, the following are out of scope for this project — not by choice, but because the underlying columns do not exist in the source data:

- Product mix and category performance analysis
- Margin or profitability analysis
- Customer segmentation (e.g. RFM analysis)
- Basket / association analysis
- Regional or location-based comparison
- Full-year seasonality and forecasting (blocked by the Date column defect in Section 3.2)

Stating these limits explicitly is a deliberate choice. Fabricating columns to populate a more conventional-looking dashboard would produce a project that is more visually complete and less truthful than the one presented here.

7. Conclusion

This project demonstrates that a raw operational export cannot be assumed clean simply because it opens as a well-formed spreadsheet. Of the eight columns supplied, three were unusable in their original form (Month, Year, Quarter), one was fully redundant (TxnDate), and the primary date field was reliable for less than 8% of records. Despite this, the underlying revenue figures were fully recoverable: the reconciliation in Section 5 confirms that the cleaned dataset accounts for every rupee in the source file. The resulting dataset supports a small number of well-founded conclusions about transaction value, and deliberately avoids overreaching into questions the data was never equipped to answer.

8. Recommended Next Steps

- Request a re-export from the source Tally system with the date field intact for all transactions, which would unlock full-year seasonality, day-of-week staffing analysis, and forecasting.
- Request item-level detail (product, quantity, cost) to enable product mix and margin analysis.
- Introduce an automated reconciliation check (subtotal-to-detail tie-out) so future monthly exports are validated on import rather than after the fact.